



10592 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Yao-Lun Yang (PI)	RIKEN Wako Institute
Dr. Zak Luka Smith (CoI) (ESA Member)	Universiteit Leiden
Dr. Scott J. Wolk (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. Pooneh Nazari (CoI) (ESA Member)	European Southern Observatory - Germany
Dr. Leslie Looney (CoI)	University of Illinois at Urbana - Champaign
Dr. Mayank Narang (CoI)	Jet Propulsion Laboratory
Dr. Neal J. Evans II (CoI)	University of Texas at Austin
Dr. Alessio Caratti o Garatti (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Capodimonte
Dr. Sam Federman (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Capodimonte
Dr. Nicole Karnath (CoI)	Space Science Institute
Dr. Ewine F. Van Dishoeck (CoI) (ESA Member)	Universiteit Leiden
Dr. Jeong-Eun Lee (CoI)	Seoul National University
Mr. Cheng-Han Hsieh (CoI)	University of Texas at Austin
Dr. Robert Gutermuth (CoI)	University of Massachusetts - Amherst
Dr. Eleonora Fiorellino (CoI) (ESA Member)	Universita di Bologna
Dr. Tom Megeath (CoI) (US Admin CoI)	University of Toledo
Dr. James Muzerolle Page (CoI)	Space Telescope Science Institute
Prof. Alexander Tielens (CoI)	University of Maryland

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSpec MSA				
	10	HEFE v5 part 1	NIRSpec MultiObject Spectroscopy	(1) mpt_cat_cy5_v2_part1
	11	HEFE v5 part 2	NIRSpec MultiObject Spectroscopy	(2) mpt_cat_cy5_v2_part2
	12	HEFE v5 part 3	NIRSpec MultiObject Spectroscopy	(3) mpt_cat_cy5_v2_part3
	13	HEFE v5 part 4	NIRSpec MultiObject Spectroscopy	(4) mpt_cat_cy5_v2_part4

ABSTRACT

Ice plays a dominant role in the chemical evolution from molecular clouds to protostars and planets. At the early stage of ice evolution, H₂O, CO, and CO₂ are the main ice species. Although these are conventionally pictured as separate layers, some JWST observations suggested that they are in mixed ice mantles, which have profound implications on the formation of organic species on ice. Furthermore, ice measurements have been focusing on lower density clouds and individual protostars, probing the initial ice compositions and the thermal processing due to protostellar accretion. Observations of regions forming more massive stars are needed to constrain the effects of UV irradiation on ice compositions. This proposal aims to use JWST to map the distribution of major ice species, H₂O, CO, and CO₂, in the integral shape filament in Orion A, probing the ice evolution over a range of extinction and UV irradiation in a dense star-forming filament. Using NIRSpec MOS, we can efficiently observe ~80 background stars and YSOs at A_v from 3-70. This filament is one of the most studied star-forming clouds with complementary data from near-IR to sub-mm/mm, which is also the focus of a recent JWST Large Program. The ice distributions in this region will survey the ice evolution at its early stage and complete the missing piece in understanding the UV processing on ice.

OBSERVING DESCRIPTION

The proposed observations use NIRSpec MOS to obtain 2-5 μ m spectra from ~80 sightlines toward background stars and YSOs in the Integral Shape Filament. The MOS observations consist of four MSA fields, covering the field of view of existing NIRCам images. Using the sources identified from archival NIRCам images, we configured each MSA setup to prioritize brighter background sources. Two gratings, G235M and G395M are requested to cover the spectral range necessary to measure H₂O, CO, and CO₂ ice.

Proposal 10592 - Targets - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	mpt_cat_cy5_v2_part1	RA: 05 35 29.1727 (83.8715529d) Dec: -05 01 0.23 (-5.01673d) Equinox: J2000		
	Comments: Description=				
	(2)	mpt_cat_cy5_v2_part2	RA: 05 35 25.5806 (83.8565858d) Dec: -05 05 15.68 (-5.08769d) Equinox: J2000		
	Comments: Description=				
	(3)	mpt_cat_cy5_v2_part3	RA: 05 35 28.5158 (83.8688158d) Dec: -05 08 16.11 (-5.13781d) Equinox: J2000		
	Comments: Description=				
	(4)	mpt_cat_cy5_v2_part4	RA: 05 35 23.6201 (83.8484171d) Dec: -05 12 29.22 (-5.20812d) Equinox: J2000		
	Comments: Description=				

Proposal 10592 - Observation 10 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Observation	Proposal 10592, Observation 10: HEFE v5 part 1										Tue May 19 13:00:29 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpect MultiObject Spectroscopy											
Diagnostics	(HEFE v5 part 1 (Obs 10)) Warning (Form): NIRSpect MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
	(HEFE v5 part 1 (Obs 10)) Warning (Form): NIRSpect MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	mpt_cat_cy5_v2_part1	RA: 05 35 29.1727 (83.8715529d) Dec: -05 01 0.23 (-5.01673d) Equinox: J2000								
Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ and New MSA Config	MSA Center	mpt_cat_cy5_v2_part1 (114 sources)	mpt_cat_cy5_v2_part1 (114 sources)	jwst-nirspec-mr	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	c1	NRSIRS2RAPID	8	1	1	131.3				
	2	ALLCLOSED	NRSIRS2RAPID	8	1	1	131.3				

Proposal 10592 - Observation 10 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	3 (G235M/F170LP)	c1	3 Shutter Slitlet	83.872171875 Degrees - 5.0161338888888 74 Degrees	74.999947562348 04			3	3	2232.1
	2	1 (G395M/F290LP)	c1	3 Shutter Slitlet	83.872171875 Degrees - 5.0161338888888 74 Degrees	74.999947562348 04			3	15	10066.334
	3	4 (G235M/F170LP)	ALLCLOSED	3 Shutter Slitlet	83.872171875 Degrees - 5.0161338888888 74 Degrees	74.999947562348 04			3	3	700.267
	4	2 (G395M/F290LP)	ALLCLOSED	3 Shutter Slitlet	83.872171875 Degrees - 5.0161338888888 74 Degrees	74.999947562348 04			3	3	919.1
Special Requirements	Aperture PA Range 70 to 80 Degrees (V3 291.4254303 to 301.4254303) MSA Planned Aperture PA 75.0000 to 75.0000 Degrees (V3 296.4254303 to 296.4254303)										

Proposal 10592 - Observation 11 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Observation	Proposal 10592, Observation 11: HEFE v5 part 2										Tue May 19 13:00:29 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(HEFE v5 part 2 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
	(HEFE v5 part 2 (Obs 11)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	mpt_cat_cy5_v2_part2	RA: 05 35 25.5806 (83.8565858d) Dec: -05 05 15.68 (-5.08769d) Equinox: J2000								
Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		After Target ACQ and New MSA Config	MSA Center	mpt_cat_cy5_v2_part2 (92 sources)	mpt_cat_cy5_v2_part2 (92 sources)		jwst-nirspec-mr		1.5
Reference Stars											
Confirmation	#	Confirmation Type			Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time
	1	c1			NRSIRS2RAPID	8	1		1		131.3
	2	ALLCLOSED			NRSIRS2RAPID	8	1		1		131.3

Proposal 10592 - Observation 11 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	3 (G235M/F170LP)	c1	3 Shutter Slitlet	83.857669125 Degrees - 5.0877447222222 26 Degrees	74.999904838228 33			3	3	2232.1
	2	1 (G395M/F290LP)	c1	3 Shutter Slitlet	83.857669125 Degrees - 5.0877447222222 26 Degrees	74.999904838228 33			3	15	10066.334
	3	4 (G235M/F170LP)	ALLCLOSED	3 Shutter Slitlet	83.857669125 Degrees - 5.0877447222222 26 Degrees	74.999904838228 33			3	3	700.267
	4	2 (G395M/F290LP)	ALLCLOSED	3 Shutter Slitlet	83.857669125 Degrees - 5.0877447222222 26 Degrees	74.999904838228 33			3	3	919.1
Special Requirements	Aperture PA Range 70 to 80 Degrees (V3 291.4254303 to 301.4254303) MSA Planned Aperture PA 75.0000 to 75.0000 Degrees (V3 296.4254303 to 296.4254303)										

Proposal 10592 - Observation 12 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Observation	Proposal 10592, Observation 12: HEFE v5 part 3										Tue May 19 13:00:29 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(HEFE v5 part 3 (Obs 12)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
	(HEFE v5 part 3 (Obs 12)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	mpt_cat_cy5_v2_part3	RA: 05 35 28.5158 (83.8688158d) Dec: -05 08 16.11 (-5.13781d) Equinox: J2000								
Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ and New MSA Config	MSA Center	mpt_cat_cy5_v2_part3 (84 sources)	mpt_cat_cy5_v2_part3 (84 sources)	jwst-nirspec-mr	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	c1	NRSIRS2RAPID	8	1	1	131.3				
	2	ALLCLOSED	NRSIRS2RAPID	8	1	1	131.3				

Proposal 10592 - Observation 12 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	3 (G235M/F170LP)	c1	3 Shutter Slitlet	83.868829874999 99 Degrees - 5.13904111111111 12 Degrees	74.999996480103 08			3	3	2232.1
	2	1 (G395M/F290LP)	c1	3 Shutter Slitlet	83.868829874999 99 Degrees - 5.13904111111111 12 Degrees	74.999996480103 08			3	15	10066.334
	3	4 (G235M/F170LP)	ALLCLOSED	3 Shutter Slitlet	83.868829874999 99 Degrees - 5.13904111111111 12 Degrees	74.999996480103 08			3	3	700.267
	4	2 (G395M/F290LP)	ALLCLOSED	3 Shutter Slitlet	83.868829874999 99 Degrees - 5.13904111111111 12 Degrees	74.999996480103 08			3	3	919.1
Special Requirements	Aperture PA Range 70 to 80 Degrees (V3 291.4254303 to 301.4254303) MSA Planned Aperture PA 75.0000 to 75.0000 Degrees (V3 296.4254303 to 296.4254303)										

Proposal 10592 - Observation 13 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Observation	Proposal 10592, Observation 13: HEFE v5 part 4										Tue May 19 13:00:29 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRSpec MultiObject Spectroscopy											
Diagnostics	(HEFE v5 part 4 (Obs 13)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
	(HEFE v5 part 4 (Obs 13)) Warning (Form): NIRSpec MOS science exposures should not normally use the ALLOPEN or ALLCLOSED MSA configurations.										
(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	mpt_cat_cy5_v2_part4	RA: 05 35 23.6201 (83.8484171d) Dec: -05 12 29.22 (-5.20812d) Equinox: J2000								
Comments: Description=[]											
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	After Target ACQ and New MSA Config	MSA Center	mpt_cat_cy5_v2_part4 (87 sources)	mpt_cat_cy5_v2_part4 (87 sources)	jwst-nirspec-mr	1.5			
Reference Stars											
Confirmation	#	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	c1	NRSIRS2RAPID	8	1	1	131.3				
	2	ALLCLOSED	NRSIRS2RAPID	8	1	1	131.3				

Proposal 10592 - Observation 13 - Early evolution of ice and UV processing in Orion's Integral Shaped Filament

Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	3 (G235M/F170LP)	c1	3 Shutter Slitlet	83.848325666666 65 Degrees - 5.2083136111111 23 Degrees	75.000007838280 85			3	3	2232.1
	2	1 (G395M/F290LP)	c1	3 Shutter Slitlet	83.848325666666 65 Degrees - 5.2083136111111 23 Degrees	75.000007838280 85			3	15	10066.334
	3	4 (G235M/F170LP)	ALLCLOSED	3 Shutter Slitlet	83.848325666666 65 Degrees - 5.2083136111111 23 Degrees	75.000007838280 85			3	3	700.267
	4	2 (G395M/F290LP)	ALLCLOSED	3 Shutter Slitlet	83.848325666666 65 Degrees - 5.2083136111111 23 Degrees	75.000007838280 85			3	3	919.1
Special Requirements	Aperture PA Range 70 to 80 Degrees (V3 291.4254303 to 301.4254303) MSA Planned Aperture PA 75.0000 to 75.0000 Degrees (V3 296.4254303 to 296.4254303)										